




FUNDAMENTALS IN RFID

Part 1

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PART 1

INTRODUCTION

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Materials distributed

- ◆ Paper “Fundamentals in RFID”
- ◆ Paper “Dense reader problems in RFID deployment”
- ◆ Paper “Definitions of terms used by the Adelaide Auto-ID Laboratory in Anti Counterfeiting Studies”
- ◆ Paper “ A Security and Authentication Primer”
- ◆ Presentation “Current RFID Research at Adelaide”
- ◆ Paper (later) “Answers to Exercises”

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How I will present

- ◆ We have 50 minute sessions with 10 minutes breaks (plus lunch break)
- ◆ Regard the papers as background material
- ◆ Hope they have been distributed and read
- ◆ Have merged that content into two major presentations (Fundamentals parts 1 and 2) for days 1 and 2.
- ◆ In the process did edit the content a little.

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Good news



Heckling is encouraged

Confession

Hand waving will occur

Warning

Questions will be asked

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Outline: 1



- ◆ Introduction
- ◆ General RFID concepts
- ◆ Auto-ID and EPCglobal concepts
- ◆ Electromagnetic Fields and propagation
- ◆ Design issues in RFID systems
- ◆ Electromagnetic coupling in the far field
- ◆ Electromagnetic coupling in the near field
- ◆ Coupling volume theory and homework

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Outline: 2



- ◆ Answers to homework
- ◆ RFID protocols
- ◆ Dense reader problems and issues
- ◆ EPCglobal network elements and functions
- ◆ Security and authentication
- ◆ Adelaide Auto-ID research program
- ◆ Conclusions

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What I aim to convey



- ◆ The physical or mathematical principles necessary to establish the feasibility of RFID systems
- ◆ The sense of excitement of working in the area
- ◆ Some important themes will be repeated
- ◆ Hopefully no pages full of algebra

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PART 2

GENERAL RFID CONCEPTS

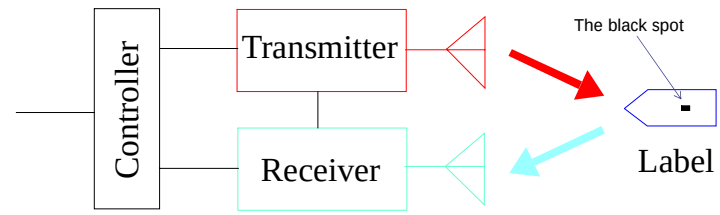
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Tag reading



Normally a very weak reply is obtained

Some application illustrations will be given shortly

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Traffic Monitoring



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Waste Collection



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EAS Gates





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Library





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Example applications



◆ What can you do with this technology ?

- Supply chain benefits
 - Reduce out of stocks, reduce inventory, speed up delivery, check freshness, track and trace, produce to demand, identify sources of diversion, identify counterfeiting, theft prediction, faster recalls
- Consumer benefits
 - Direct order from home, smart appliances, (e.g. microwave, washing machine, refrigerator), smart healthcare, assisted living
- New and less expected benefits
 - Customized products, smart recycling, checkout-less stores

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The supply chain



Global Supply Chain



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PART 3

AUTO-ID AND EPCglobal CONCEPTS

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The Auto-ID Center

- ◆ Global, industry funded research program
 - Massachusetts Institute of Technology (1999)
 - Cambridge University (2000)
 - University of Adelaide (2001-2002)
 - Japan, China, Switzerland (2003)
 - Korea (2005)
- ◆ Mission
 - Create the internet of things
 - Research for the benefit of mankind

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The Auto-ID Center Vision

- ◆ The internet of things
 - Physical objects connected via the internet
 - Simple identifying labels on objects
 - Unlimited associated data in a data base
 - Connections via an intranet or the internet
- ◆ Freely available world wide standards
- ◆ High performance protocols and software
- ◆ A scalable system not choked by expansion

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Metamorphosis

- ◆ Auto-ID Center
 - Terminated 31/10/2003
- ◆ Spawned two organisations
 - Auto-ID Labs
 - ◆ MIT, Cambridge (UK), Adelaide, Fudan (China), Keio (Japan), St Gallen/ETHZ (Switzerland) and in 2005 ICU (Korea)
 - EPCglobal





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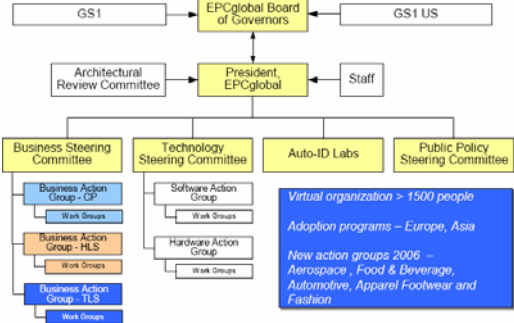

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EPCglobal Organization



Virtual organization > 1500 people
Adoption programs – Europe, Asia
New action groups 2006 – Aerospace, Food & Beverage, Automotive, Apparel Footwear and Fashion

EPCglobal
Powered by GS1

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EPCglobal Membership Mar 2006



Region	End Users	Solution Providers	Total Subscribers
1 U.S./Canada	307	159	466
2 Latin America	4	17	21
3 Europe	81	70	151
4 Africa	2	10	13
5 Asia Pacific	48	115	163
Totals	423	371	794

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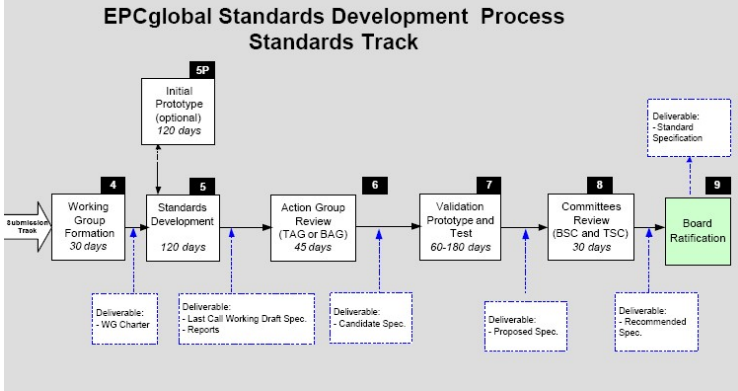
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Standards Development Process

EPCglobal Standards Development Process Standards Track



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Working Groups



- ◆ Business Steering Committee (BSC)
 - Fast Moving Consumer Goods (FMCG)
 - Healthcare and Life Sciences (HLS)
 - Transport and Logistics (TLS)
- ◆ Technical Steering Committee (TSC)
 - Hardware Action Group (HAG)
 - Software Action Group (SAG)

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FMCG – Working Groups



- ◆ Data Exchange
- ◆ European Adoption Programme (EAP)
- ◆ Pilot and Implementation (P&I)
- ◆ Reusable Transport Items (RTI)
- ◆ Strategic Planning
- ◆ Tag and Inlay Standards
- ◆ Asian Adoption Program (AAP)



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HLS – Working Groups

Slide Updated since print



- ◆ Strategy
- ◆ Policy (Complete)
- ◆ Process (Complete)
- ◆ Information
- ◆ Technology
- ◆ Research & Development (on hold)
- ◆ Track & Trace
- ◆ Medical Devices



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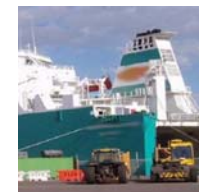
27



TLS - Working Groups



- ◆ Transportation
- ◆ “Four walls”
- ◆ Import Export Clearance
- ◆ Integration



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HAG – Working Groups



- ◆ Class 1 Generation 2 (Work completed)
- ◆ Gen 2 Testing & Certification
- ◆ Air Interface (Item Level Tagging)
 - UHF AI
 - HF AI
- ◆ Interoperability



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SAG Working Groups



- ◆ Reader Protocol => Reader Operations
- ◆ Reader Management
- ◆ Filtering and Collection (Phase 1 Complete)
- ◆ ONS (Complete)
- ◆ Security (Phase 1 Complete)
- ◆ Tag Data Translation (Complete)
- ◆ EPCIS Phase 2
- ◆ Tag Data Standards (Phase 1 Complete)
- ◆ Drug Pedigree Messaging (New)
- ◆ Tag Data and Translation Standards (New)

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Future Business Action Groups



- ◆ Aerospace and Defense
 - Airbus
 - Boeing
 - Lockheed Martin
 - NASA Space
 - U.S. Department of Defense
- ◆ Food and Beverage
 - Livestock
 - Fruits/vegetables subject to disease/infestation
- ◆ Auto
 - Early discussions (BMW, Daimler-Chrysler, Ford, GM, Hyundai, Toyota, Volkswagen)
- ◆ Apparel and Footwear



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Future Industries



- ◆ High-Tech
 - Computer & Consumer electronics
 - Industrial electronics
- ◆ Chemical
 - Petrochemicals
 - Industrial chemicals
- ◆ Defence
 - War materiel
 - Equipment maintenance
- ◆ Oil & Gas
 - Exploration



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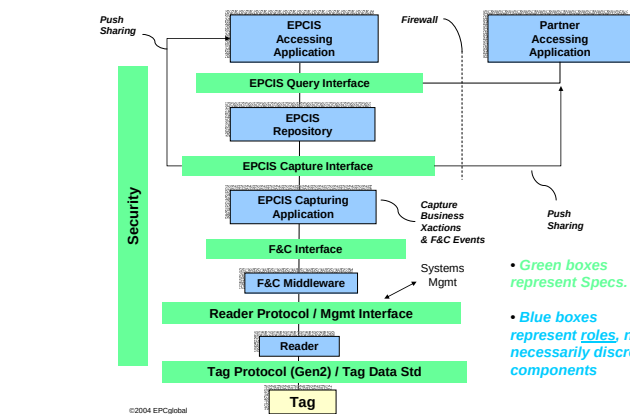



EPCglobal Network Overview

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EPCglobal network: roles and interfaces



- Green boxes represent Specs.
- Blue boxes represent roles, not necessarily discrete components

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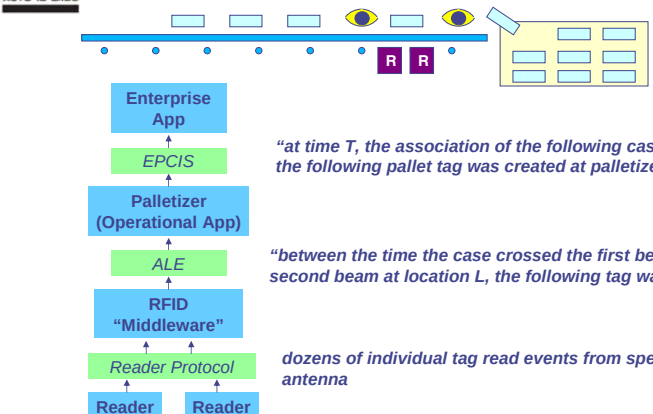



EPCIS

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EPC Event Layers

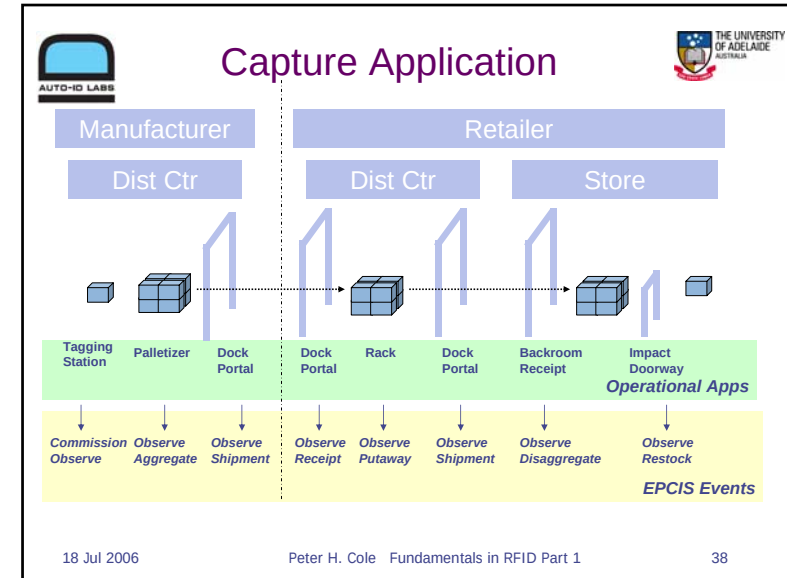
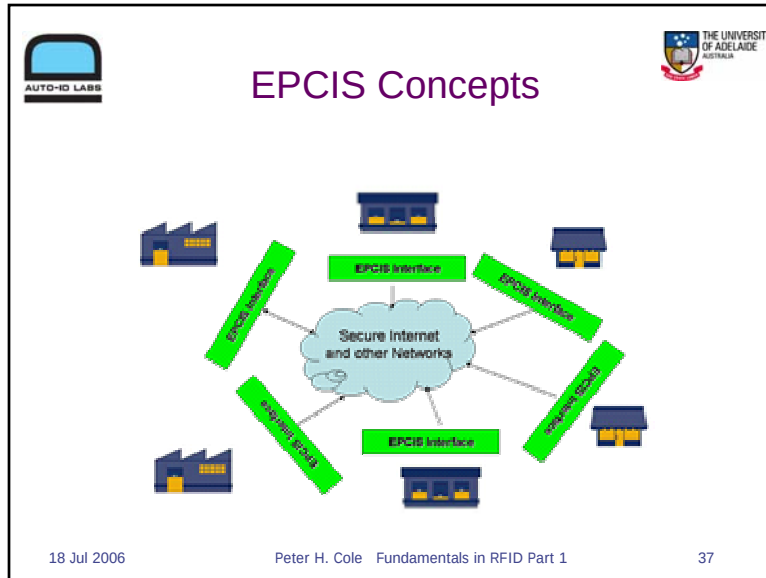


"at time T, the association of the following case tags to the following pallet tag was created at palletizer #3"

"between the time the case crossed the first beam and the second beam at location L, the following tag was read"

dozens of individual tag read events from specific antenna

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PART 4

ELECTROMAGNETIC FIELDS AND PROPAGATION

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Use of electromagnetic fields

- ◆ Coupling is via electromagnetic fields
- ◆ There is little margin for poor performance
- ◆ We must understand their properties

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The field vectors



A full theory of electrodynamics, including the effects of dielectric and magnetic materials, must be based on the four field vectors:

- Electric field vector $\mathcal{E}$
- Magnetic field vector $\mathcal{H}$
- Electric flux density vector $\mathcal{D}$
- Magnetic flux density vector $\mathcal{B}$

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Material state vectors



$\mathcal{P}$ for a dielectric material

$\mathcal{M}$ for a magnetic material

$$\mathcal{D} = \epsilon_0 \mathcal{E} + \mathcal{P}$$

$$\mathcal{B} = \mu_0 (\mathcal{H} + \mathcal{M})$$

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Laws in differential form



$$\begin{aligned}\nabla \times \mathcal{E} &= -\frac{\partial \mathcal{B}}{\partial t} \\ \nabla \times \mathcal{H} &= \mathcal{J} + \frac{\partial \mathcal{D}}{\partial t} \\ \nabla \cdot \mathcal{D} &= \rho \\ \nabla \cdot \mathcal{B} &= 0\end{aligned}$$



Vortex



Source

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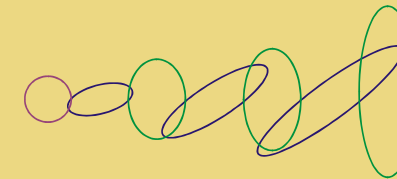
Electromagnetic propagation



Electric current creates a vortex of magnetic field

Magnetic field creates a vortex of electric field

Electric field creates a vortex of magnetic field



Propagation

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Electromagnetic waves



- ◆ They propagate with the velocity of light
 - (Light is an electromagnetic wave)
- ◆ Velocity c is 300,000,000 m/s
- ◆ Wavelength - frequency relation is $c = f\lambda$
- ◆ Propagation constant $\beta = 2\pi/\lambda$
- ◆ But not all electromagnetic fields are propagating waves; some are just local energy storage fields

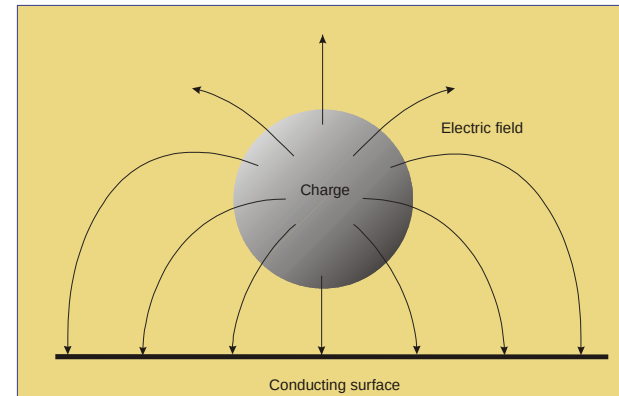
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Boundary Condition: electric field



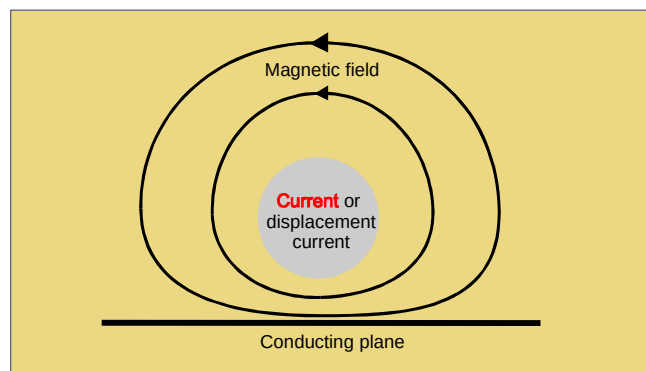
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Boundary Condition: magnetic field



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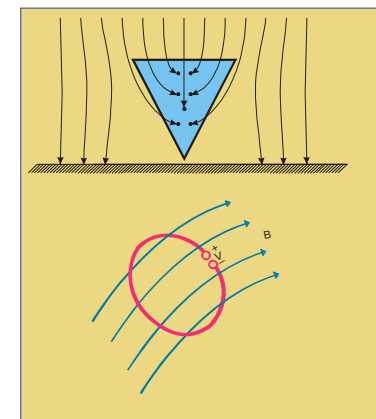


The basic laws: how they work



◆ Gauss's law

- Electric flux deposits charge
- Electric field cannot just go past a conductor, it must turn and meet it at right angles



◆ Faraday's law

- Oscillating magnetic flux induces voltage in a loop that it links

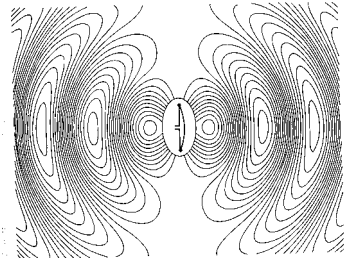
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Near and far field distributions



Electric field launched by an electric dipole

There is also a magnetic field not shown

Note the differences between near and far fields

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Fields of a Magnetic Dipole

(oh dear, I lied)



$$H_r = \frac{j\beta^3 M}{4\pi} \left(\frac{2}{(\beta r)^2} - \frac{2j}{(\beta r)^3} \right) e^{-j\beta r} \cos \theta$$

$$H_\theta = \frac{j\beta^3 M}{4\pi} \left(\frac{j}{(\beta r)} + \frac{1}{(\beta r)^2} - \frac{j}{(\beta r)^3} \right) e^{-j\beta r} \sin \theta$$

$$E_\phi = \frac{j\beta^3 M}{4\pi} \left(\frac{j}{(\beta r)} + \frac{1}{(\beta r)^2} \right) e^{-j\beta r} \sin \theta$$

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Near and far fields



- ◆ The far field is an energy propagating field
- ◆ Appropriate measure of strength is $0.5 \eta H^2$ (power flowing per unit area)
- ◆ The near field is an energy storage field
- ◆ Appropriate measure of strength is reactive power per unit volume $0.5 \omega \mu_0 H^2$
- ◆ Near field - far field boundary is $\lambda/2\pi$
- ◆ Examples 100 kHz 500m; 10 MHz 5m; 1000 MHz 50mm

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The radian sphere



At $\beta r = 1$, $r = \lambda/2\pi$, and

- The phase factor $e^{-j\beta r}$ is one radian
- Inside this sphere the near field predominates
- Outside this sphere the far field predominates

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PART 5

DESIGN ISSUES IN RFID SYSTEMS

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Issues in RFID design



- ◆ Active or passive
- ◆ Operating frequency
- ◆ Electric or magnetic fields
- ◆ Material or microelectronic

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The usual way: backscatter



- ◆ The most popular technology
- ◆ Tag contains a microcircuit and an antenna
- ◆ Tag is powered by the interrogation beam
- ◆ Frequency of that beam is chosen for good propagation
- ◆ Tag contains an internal oscillator
- ◆ Frequency of that oscillator is chosen for low power consumption

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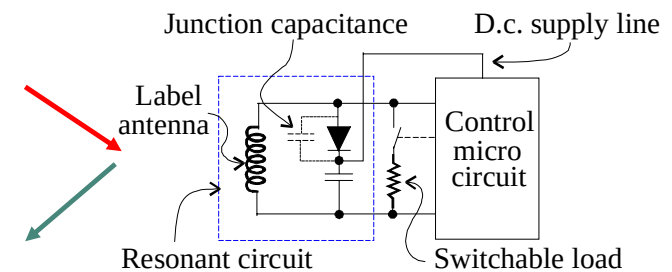
55



Microelectronic backscatter



- ◆ Concept can be applied from 10 MHz to 10,000 MHz
- ◆ Low propagation loss points to coupling using the far field
- ◆ Low power consumption requires a low frequency microcircuit
- ◆ Reply is by modulation of the interrogation frequency



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Relevant issues



- ◆ Range is determined largely by the ability to obtain sufficient rectified voltage for the label rectifier system
- ◆ High quality factor resonance becomes important in small tags
- ◆ Reply is at sidebands of the interrogation frequency

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Interesting features



- ◆ Near and far fields
- ◆ Energy storage in the near field
- ◆ Energy propagation in the far field
- ◆ Radian sphere ($r=\lambda/2\pi$) is the boundary

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Label antennas



- ◆ Magnetic field – free space
- ◆ Magnetic field against metal
- ◆ Electric field – free space
- ◆ Electric field against metal
- ◆ Electromagnetic field
- ◆ Very small antennas respond to either the electric field or the magnetic field
- ◆ Somewhat larger antennas respond to both

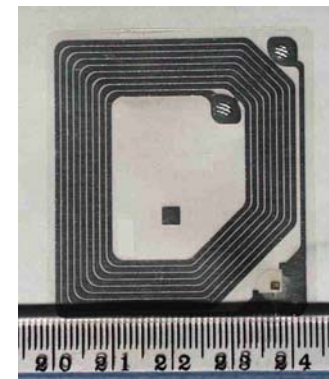
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Planar printed coil



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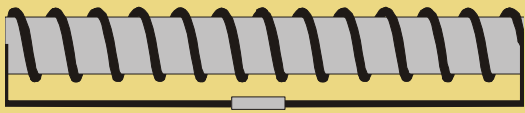
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Ferrite cored solenoid





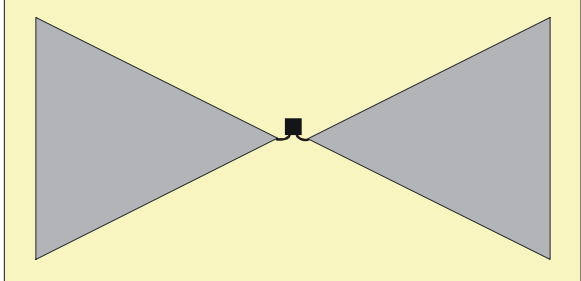
$$\mu_{er} = \frac{\mu_{ir}}{1 + N(\mu_{ir} - 1)}$$

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Electric field bow tie



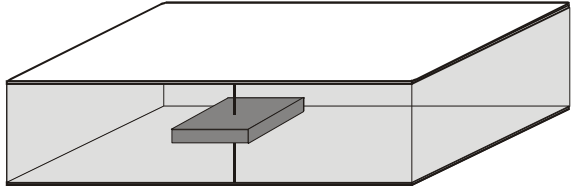


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Electric field box structure



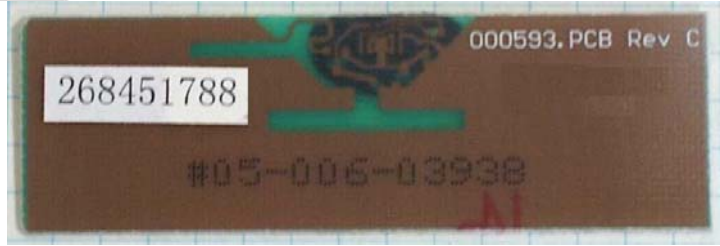


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Electromagnetic field antenna





- ◆ Dimensions are no longer a small fraction of a wave length
- ◆ Operating principles are less clear

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Field creation structures



- ◆ Near magnetic field
 - Made by current carrying loops
- ◆ Near electric field
 - Made by charged electrodes
- ◆ Far electromagnetic field
 - Made by propagation from an originally near field

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PART 6

ELECTROMAGNETIC COUPLING IN THE FAR FIELD

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Far field coupling theory: 1



$P_r = A_{er} \times \text{Power flow per unit area (defines } A_{er})$

Power flow per unit area $S_r = \frac{g_t P_t}{4\pi r^2}$ (defines g_t)

$A_{er} = \frac{g_r \lambda^2}{4\pi}$ (from Lorentz reciprocity relation)

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Far field coupling theory: 2



$$P_r = S_r A_e = \frac{g_r \lambda^2}{4\pi} S_r$$

$$\frac{P_r}{P_t} = g_r g_t \left(\frac{\lambda}{4\pi r} \right)^2$$

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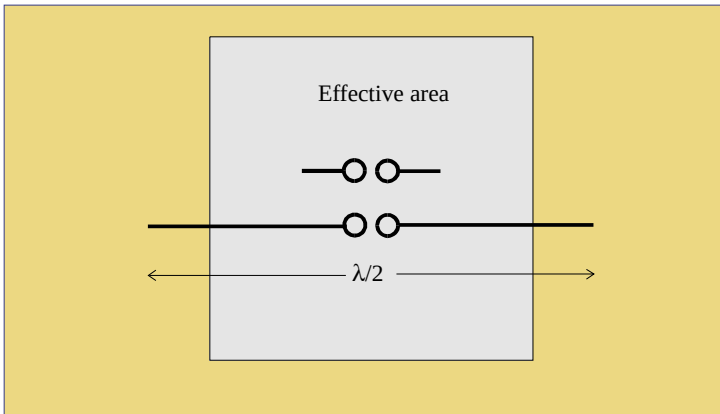
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Far field coupling theory: 3





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Far Field coupling theory: 4



- ◆ Effective area exists even for wire antennas
- ◆ It is not significantly dependent on antenna size
- ◆ That suggests we use low frequencies to get large effective areas
- ◆ That conclusion ignores the increasing radiation Q when the antenna becomes small in relation to a wave length

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PART 7

ELECTROMAGNETIC COUPLING IN THE NEAR FIELD

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Near field coupling: 1



- ◆ None of the far field concepts apply in the near field
- ◆ We want a calculation method that is based on the physics in play
- ◆ What the near field does is store energy
- ◆ We need a power-like measure of the stored energy
- ◆ That measure is reactive power per unit volume
- ◆ We want to separate the effects of geometry and quality factor

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Near field coupling: 2



- ◆ There are quite similar electric field and magnetic field versions of the theory.
- ◆ We will concentrate first on the magnetic field version.

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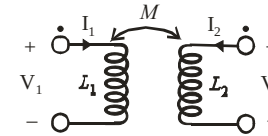
73



Near field coupling : 3

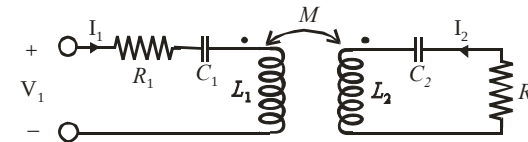


- ◆ In the near field we just have coupled coils



$$V_1 = j\omega L_1 I_1 + j\omega M I_2$$

$$V_2 = j\omega M I_1 + j\omega L_2 I_2$$



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74



Near field coupling: 4



Simple result for weakly coupled coils

$$\frac{P_2}{P_1} = k^2 Q_1 Q_2 \text{ where } k = \frac{M}{\sqrt{L_1 L_2}}$$

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PART 8

COUPLING VOLUME THEORY AND
HOMEWORK

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Coupling volume theory: 1



- ◆ Coefficient of coupling k depends on the coil sizes and positions
- ◆ Quality factor depends on materials
- ◆ We want to separate these effects
- ◆ We want to obtain for k^2 a product of factors depending separately on the interrogator coil geometry and the tag coil geometry
- ◆ The power-like measure driving field is the reactive power density per unit volume at the tag position

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Coupling volume theory: 2



$$V_c = \frac{\left[\begin{array}{c} \text{Reactive power flowing in the untuned label coil} \\ \text{when it is short circuited} \end{array} \right]}{\left[\begin{array}{c} \text{Volume density of reactive power created by} \\ \text{the interrogator at the label position} \end{array} \right]}$$

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Coupling volume theory: 3



$$V_d = \frac{\left[\begin{array}{c} \text{Reactive power flowing in the inductor} \\ \text{of the interrogator field creation coil} \end{array} \right]}{\left[\begin{array}{c} \text{Volume density of reactive power created by} \\ \text{the interrogator at the label position} \end{array} \right]}$$

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Coupling volume theory: 4



- ◆ When we tune the coil, the current increases by a factor Q , and the reactive power in the coil increases by a factor Q^2 .
- ◆ The real power needed to sustain the resonance is that increased reactive power divided by Q .
- ◆ We need to take into account the relation of power delivered to the interrogator coil to its total circulating reactive power.

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80



Coupling volume theory: 5



$$\frac{P_2}{P_1} = \frac{V_c}{V_d} Q_1 Q_2$$

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81



Coupling volume theory: 6



- ◆ For some antennas, e.g. long air cored solenoids, coupling volume is equal to the volume of the solenoid.
- ◆ But coupling volume exists even for planar antennas, e.g. for a planar coil of flux collecting area A and self inductance L the coupling volume is $V_c = \mu_0 A^2 / L$

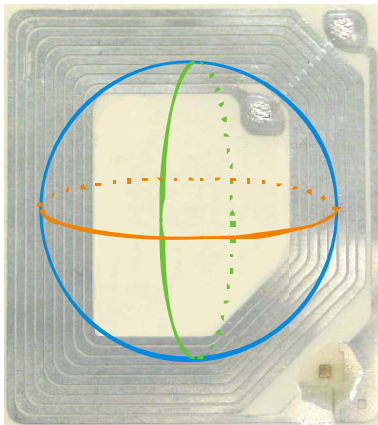
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82



Coupling volume theory: 7



18 Jul 2006

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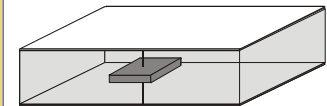
83



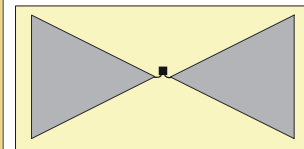
Some electric field examples



For a pair of air cored parallel electrodes $V_c =$ volume enclosed
A dielectric if present reduces the coupling volume by ϵ_r



For a bow tie, there is no physical volume but there is a V_c depending on the self capacitance and the electric flux collecting area



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84



Application in the far field



- ◆ Small antennas will respond in the same way to near and far field excitation.
- ◆ They can be analysed by either effective area or coupling volume concepts, but if the loss resistance is much greater than the radiation resistance so that it greatly impedes the extraction of power, coupling volume concepts are better.

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85



Near and far field excitation



In the far field

$$W_v = \beta S_r$$

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86



Significant conclusions



- ◆ Coupling volumes for well shaped planar electric and magnetic field labels are size dependent and similar

$$\text{Magnetic } V_c = \frac{L^3}{2} \quad \text{Electric } V_c = \frac{2L^3}{3}$$

- ◆ Radiation quality factors for both types of label formed within a square of side L are size dependent and similar

$$\text{Magnetic } Q_r = \frac{40}{(\beta L)^3} \quad \text{Electric } Q_r = \frac{13}{(\beta L)^3}$$

- ◆ These are calculated results for sensibly shaped antennas

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87



Optimum operating frequency



The optimum frequency for operation of an RFID system in the far field is the lowest frequency for which a reasonable match to the radiation resistance of the label antenna can be achieved, at the allowed size of label, without the label or matching element losses intruding.

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88



Homework



- ◆ Perform the exercises in Appendix D of the paper “Fundamentals in RFID”
- ◆ The formulae in Appendix C of that paper may be useful
- ◆ The answers will be discussed in the next day of the course.

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89



Thank you

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90